

MH3100 Real Analysis I

Midterm Practice 2 (Solutions)

AY 2026/2027, Semester 1 – QRS original

Quantitative Research Society @NTU

October 1, 2026

TIME ALLOWED: 60 minutes.

Instructions to Students

1. This test contains **FOUR (4)** questions.
2. Answer **all** questions. The marks for each question are indicated below.
3. Write your answers in the box provided. Use the **facing page** for more space.
4. This is a **CLOSED BOOK** test.
5. You may use calculators. You may cite any results proved in the lectures and tutorials, **unless you are asked by the question to prove the result.**

Question	1	2	3	4	Total
Total Marks	6	6	6	6	24
Awarded Marks					

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Question 1**(6 Marks)**

Is each of the following statement true or false? You do NOT need to justify your answer.

- (a) (2 marks) If $\{a_n\}$ is a bounded sequence such that every convergent subsequence of $\{a_n\}$ converges to the same real number L , then $\{a_n\}$ converges to L .
- (b) (2 marks) There is a monotone sequence that has a convergent subsequence but is itself divergent.
- (c) (2 marks) If $\{a_n\}$ and $\{b_n\}$ are Cauchy sequences with $b_n \neq 0$ for every $n \in \mathbb{N}$, then $\left\{\frac{a_n}{b_n}\right\}$ is a Cauchy sequence.

Solution

- (a) ☐ True

Proof. Suppose $\{a_n\}$ does not converge to L . Then for some $\varepsilon > 0$, $|a_n - L| \geq \varepsilon$ for infinitely many n (def. of limit). $\therefore$ infinitely many n satisfy $a_n - L \geq \varepsilon$, or infinitely many satisfy $L - a_n \geq \varepsilon$. $\therefore$ there is a bounded subsequence $\{b_k\}$ of $\{a_n\}$ with $b_k - L \geq \varepsilon$ for all k , or $L - b_k \geq \varepsilon$ for all k . (BW) gives a subsequence of $\{b_k\}$, hence of $\{a_n\}$, with limit L' . Every term of that subsequence has $b_k - L \geq \varepsilon$ (resp. $L - b_k \geq \varepsilon$).

$$\implies L' - L \geq \varepsilon \quad (\text{resp. } L - L' \geq \varepsilon) \quad (\text{OLT})$$

$\therefore L' \neq L$, contradicting the hypothesis. $\therefore L = \lim a_n$. $\square$

- (b) ☐ False

Proof. (Monotone Convergence Lemma) A monotone sequence with a convergent subsequence converges. $\therefore$ no monotone sequence has a convergent subsequence and diverges. $\square$

- (c) ☐ False

Proof. Let $a_n := 1$, $b_n := \frac{1}{n} \neq 0$; both converge, hence are Cauchy (Cauchy Criterion). $\frac{a_n}{b_n} = n$ is unbounded $\implies$ divergent (convergent $\implies$ bounded) $\implies$ not Cauchy (Cauchy Criterion). $\square$

Marking scheme (6 marks)

QRS suggested marking scheme.

- (a) **(2 marks)** 2 marks for ☐ True; 0 marks for ☐ False. No justification is required on the actual exam.
- (b) **(2 marks)** 2 marks for ☐ False; 0 marks for ☐ True.
- (c) **(2 marks)** 2 marks for ☐ False; 0 marks for ☐ True.

Typical deductions (if justification is examined, e.g. in a tutorial recap): in (a), asserting the result without reference to a compactness-type tool (Bolzano-Weierstrass); in (c), attempting a "proof" instead of noticing a quotient can be unbounded; confusing Cauchy with merely bounded.

Question 2**(6 Marks)**

Is each of the following statement true or false? You do NOT need to justify your answer.

- (a) (2 marks) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_n^2$ converges.
 (b) (2 marks) Every finite subset of $\mathbb{R}$ is a closed set.
 (c) (2 marks) $\overline{\mathbb{Q} \cap [0, 1]} = [0, 1]$.

Solution

- (a) False

Proof. Let $a_n := \frac{(-1)^n}{\sqrt{n}}$. $\{1/\sqrt{n}\}$ decreases with limit 0 $\therefore \sum_{n=1}^{\infty} a_n$ converges (AST).
 $a_n^2 = \frac{1}{n}$, and for every $m \in \mathbb{N}$ the partial sums of $\sum_{n=1}^{\infty} 1/n$ satisfy

$$s_{2m} - s_m = \sum_{k=m+1}^{2m} \frac{1}{k} \geq m \cdot \frac{1}{2m} = \frac{1}{2}.$$

$\therefore \sum_{n=1}^{\infty} a_n^2$ fails the Cauchy Criterion (with $\varepsilon = \frac{1}{2}$), so it diverges. □

- (b) True

Proof. Let $F \subseteq \mathbb{R}$ be finite and $a \in \mathbb{R}$. If $F \setminus \{a\} = \emptyset$, no $x \in F \setminus \{a\}$ exists, so a is not a limit point of F . Otherwise $d := \min_{x \in F \setminus \{a\}} |x - a| > 0$ and $|x - a| \geq d$ for all $x \in F \setminus \{a\}$. $\therefore$ no $x \in F \setminus \{a\}$ has $|x - a| < d$, i.e. the definition of limit point fails for $\varepsilon = d$, so a is not a limit point of F . $\therefore F$ has no limit points, so F contains all of them: F is closed. □

- (c) True

Proof. **$[0, 1]$ is closed.** Let a be a limit point of $[0, 1]$. There is $\{x_n\} \subseteq [0, 1] \setminus \{a\}$ with $a = \lim x_n$ (Sequential Criterion for Limit Points). $0 \leq x_n \leq 1$ for all $n \implies 0 \leq a \leq 1$ (OLT). $\therefore a \in [0, 1]$, so $[0, 1]$ contains all its limit points: $[0, 1]$ is closed.

($\subseteq$) Let $a \in \overline{\mathbb{Q} \cap [0, 1]}$. Then for every $\varepsilon > 0$ some $x \in \mathbb{Q} \cap [0, 1] \subseteq [0, 1]$ has $|x - a| < \varepsilon$ (ε -Criterion for Closures). $\therefore a \in \overline{[0, 1]} = [0, 1]$ (Closure Criterion, $[0, 1]$ closed).

($\supseteq$) Let $x \in [0, 1]$ and $S_n := \{k \in \mathbb{N}_0 : k \leq nx\}$ for $n \in \mathbb{N}$. $0 \in S_n \subseteq \{0, \dots, n\}$ because $0 \leq x \leq 1$. $\therefore k_n := \max S_n$ exists, $0 \leq k_n \leq n$, $q_n := k_n/n \in \mathbb{Q} \cap [0, 1]$. $k_n \leq nx < k_n + 1$ because $k_n + 1 \notin S_n$.

$$0 \leq nx - k_n < 1 \implies |q_n - x| = \frac{|nx - k_n|}{n} < \frac{1}{n}$$

Let $\varepsilon > 0$; choose $N \in \mathbb{N}$ with $\frac{1}{N} < \varepsilon$ (Archimedean). For all $n \geq N$: $|q_n - x| < \frac{1}{n} \leq \frac{1}{N} < \varepsilon$. $\therefore x = \lim q_n$, so $x \in \overline{\mathbb{Q} \cap [0, 1]}$ (Sequential Criterion for Closures).

$\therefore \overline{\mathbb{Q} \cap [0, 1]} = [0, 1]$. □

Marking scheme (6 marks)

QRS suggested marking scheme.

- (a) **(2 marks)** 2 marks for ☐ False; 0 marks for ☐ True.
- (b) **(2 marks)** 2 marks for ☐ True; 0 marks for ☐ False.
- (c) **(2 marks)** 2 marks for ☐ True; 0 marks for ☐ False.

Typical deductions (if justification is examined): in (a), assuming convergence of $\sum_{n=1}^{\infty} a_n$ is absolute; in (b), confusing "closed" with "bounded" or "compact"; in (c), forgetting to prove the reverse inclusion (that the closure cannot exceed $[0, 1]$).

Question 3

(6 Marks)

In this question, A is a nonempty set of real numbers that is bounded below, and $c > 0$. Write down the definition of $s = \inf A$, and use the definition to show that

$$\inf(cA) = c \inf A, \quad \text{where } cA := \{ca : a \in A\}.$$

Do not use the ε -Criterion for Suprema and Infima.

Solution

Method 1: the two defining properties

Definition. $s = \inf A$ means: (1) $a \geq s$ for every $a \in A$; (2) $s \geq L$ for every lower bound L of A .

Proof. Let $s = \inf A$. **Step 1.** Let $a \in A$. $a \geq s \implies ca \geq cs$ ($c > 0$). $\therefore cs$ is a lower bound of cA . **Step 2.** Let L be a lower bound of cA . For every $a \in A$: $ca \geq L \implies a \geq L/c$ ($c > 0$). $\therefore L/c$ is a lower bound of A , so $s \geq L/c$ (def. of $\inf A$), i.e. $cs \geq L$ ($c > 0$). L arbitrary $\therefore cs \geq L$ for every lower bound L of cA . Steps 1, 2 $\therefore \inf(cA) = cs = c \inf A$. $\square$

Marking scheme (6 marks)

QRS suggested marking scheme.

- **2 marks:** definition of $s = \inf A$ stated correctly, with both clauses (lower bound; not less than any lower bound) and no quantifier-order errors.
- **3 marks:** complete proof of both Step 1 (1 mark: cs is a lower bound for cA) and Step 2 (2 marks: cs is the greatest lower bound of cA , including the division step and the correct use of $s = \inf A$).
- **1 mark:** the three writing expectations are met throughout (complete sentences; related statements joined by connectives; inference marked with "therefore/hence/so").

Typical deductions: omitting Step 2 entirely; using the ε -Criterion for Suprema and Infima despite the prohibition; silently assuming $c > 0$ preserves inequality direction without stating it; sign errors when dividing by c .

Question 4

(6 Marks)

In this question, $\{a_n\}$ is a sequence of nonnegative real numbers. Write down the definition of $a = \lim a_n$, and use the definition to show that if $a = \lim a_n$, then $\sqrt{a} = \lim \sqrt{a_n}$.

Do not use the Algebraic Limit Theorem.

Solution

Method 1: rationalisation, cases $a = 0$ and $a > 0$

Definition. $a = \lim a_n$ means: for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ with $|a_n - a| < \varepsilon$ for all $n \geq N$.

Proof. $a_n \geq 0$ for all $n \implies a \geq 0$ (OLT), so $\sqrt{a}$ exists. **Case $a = 0$.** Let $\varepsilon > 0$; choose N with $|a_n - 0| < \varepsilon^2$ for all $n \geq N$ (def. of limit). $\therefore$ for $n \geq N$: $0 \leq a_n < \varepsilon^2 \implies |\sqrt{a_n} - 0| < \varepsilon$. $\therefore 0 = \lim \sqrt{a_n}$.

Case $a > 0$. Let $\varepsilon > 0$; choose N with $|a_n - a| < \varepsilon\sqrt{a}$ for all $n \geq N$ (def. of limit, $\varepsilon\sqrt{a} > 0$). For all $n \geq N$:

$$\begin{aligned} |\sqrt{a_n} - \sqrt{a}| &= \frac{|(\sqrt{a_n} - \sqrt{a})(\sqrt{a_n} + \sqrt{a})|}{\sqrt{a_n} + \sqrt{a}} && \text{(rationalise)} \\ &= \frac{|a_n - a|}{\sqrt{a_n} + \sqrt{a}} \\ &\leq \frac{|a_n - a|}{\sqrt{a}} < \varepsilon && (\sqrt{a_n} + \sqrt{a} \geq \sqrt{a} > 0) \end{aligned}$$

$\therefore \sqrt{a} = \lim \sqrt{a_n}$ in both cases. □

Method 2: square-root difference inequality

Proof. $a \geq 0$ (OLT). Let $x \geq y \geq 0$: $(\sqrt{y} + \sqrt{x-y})^2 = x + 2\sqrt{y(x-y)} \geq x \implies 0 \leq \sqrt{x} - \sqrt{y} \leq \sqrt{x-y}$. $\therefore |\sqrt{x} - \sqrt{y}| \leq \sqrt{|x-y|}$ for all $x, y \geq 0$ (swap x, y if $x < y$). Let $\varepsilon > 0$; choose N with $|a_n - a| < \varepsilon^2$ for all $n \geq N$ (def. of limit). $\therefore$ for $n \geq N$: $|\sqrt{a_n} - \sqrt{a}| \leq \sqrt{|a_n - a|} < \varepsilon$. $\therefore \sqrt{a} = \lim \sqrt{a_n}$. □

Marking scheme (6 marks)

QRS suggested marking scheme.

- **2 marks:** definition of $a = \lim a_n$ stated correctly, with the quantifiers in the correct order (N may depend on ε , not vice versa).
- **1 mark:** correctly establishes $a \geq 0$ via the Order Limit Theorem before defining $\sqrt{a}$.
- **2 marks:** correct treatment of both cases ($a = 0$ and $a > 0$), including the rationalisation identity and a valid choice of N for the given ε .
- **1 mark:** the three writing expectations are met throughout (complete sentences; related statements joined by connectives; inference marked with "therefore/hence/so").

Typical deductions: omitting the case $a = 0$; using the Algebraic Limit Theorem or citing "continuity of $\sqrt{\cdot}$ " despite the prohibition; dividing by $\sqrt{a}$ without justifying $a > 0$ first; choosing N depending on n .